

ABAQUS report

REPCE

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I) Introduction

I.1) Introduction

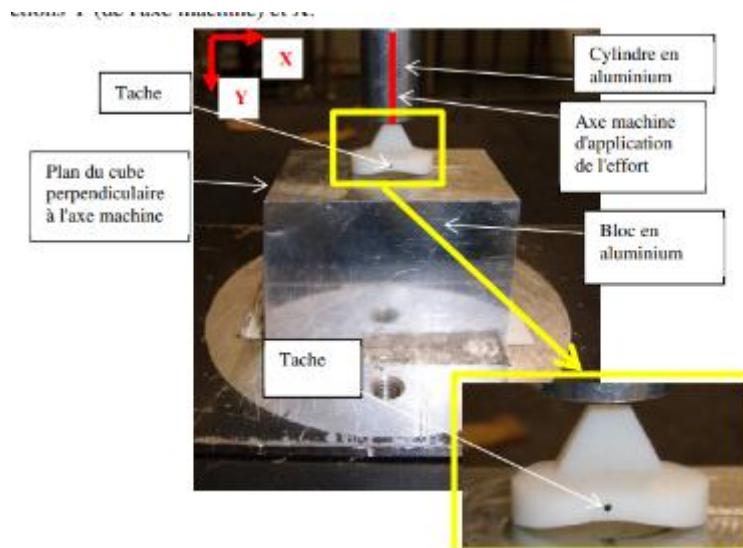


Figure 1: Expérimentation sur implant glénoïdien à quille

The present report describes the implementation and validation of a finite element model (Abaqus) of a glenoid implant with a keel, made of photopolymer resin (Zortrax Resin Basic). The objective is to build a simple and reproducible digital model, to subject it to the same loading conditions as the experimental test (axial loading via a cylinder on the flat face of the keel, support on the edge of the plate, speed of 1 mm min⁻¹) and to compare the simulated displacement with the experimentally measured displacement (0.228 mm in Y for a load of 150 N). The model is deliberately linear and elastic, which is adapted to the studied field of solicitations (small deformations under 200 N) and allows isolating the effect of geometric choices, mesh size and boundary conditions on the mechanical response.

I.2) Objective

During this test, the experiment informs us that a displacement of 0.228mm occurs when applying a force of 150N. Thus, the objective would be to establish a finite element model that perfectly illustrates this situation.

II) Model ABAQUS

II.1) Justification of the properties of the material used for our prosthesis

The material of the implant, Zortrax Resin Basic, is modeled as an isotropic and linearly elastic material, which is justified by the low applied solicitations (200 N maximum) inducing small deformations. Young's modulus was set at 2.1 GPa, the average value of the range provided by the datasheet (1.78 - 2.38 GPa), in order to obtain a representative rigidity without overestimating the stiffness of the material. The Poisson coefficient was chosen to be equal to 0.38, a typical value for photopolymer resins, reflecting a slightly compressible behavior. Thus this choice remains coherent with the material of our glenoidian component.

Finally, the density was taken at 1.09 g/cm³ (that is 1.09 10⁻³ ton/mm³ in the N-mm-MPa system) according to the datasheet. But this data does not interest us in the case of a linear elastic model

These choices guarantee a simple, coherent and sufficient model to describe the behavior of the material within the framework of a linear static analysis.

II.2) Choice of the type of force applied to the glenoid component

In the experiment, the load is applied by the plane face of an aluminum cylinder that exerts a compressive axial force on the upper surface of the keel, up to a maximum force of 200 N. In order to reproduce this strain in the finite element model, the corresponding face was subjected to a uniform pressure equivalent to a total load of 150 N, a value used to compare the simulated displacement with the available experimental result (0.228 mm at 150 N). This choice allows to accurately represent the distributed contact between the cylinder and the keel while avoiding local singularities associated with the application of a point force.

II.3) Choice of boundary conditions

The validation of the finite element model required an iterative approach, mainly focused on the correct definition of boundary conditions (LC) so that they faithfully represent the experimental setup.

For all the tests in this section, the model was meshed with tetrahedral elements with quadratic geometry (order 2). This choice is adapted to the complex geometry of the implant. A first definition of the size of elements (seeding) was applied by setting a minimum size of 0.5 and a maximum size of 1 on the main surfaces and the rest of the geometry. We carried out a control of the elements by regions rather than on the overall set of the glenoid component.

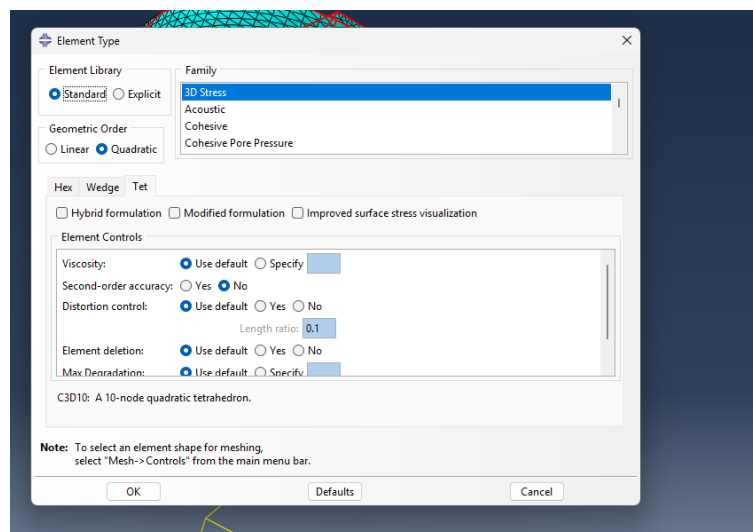


Figure II 1: Configuration of the element type

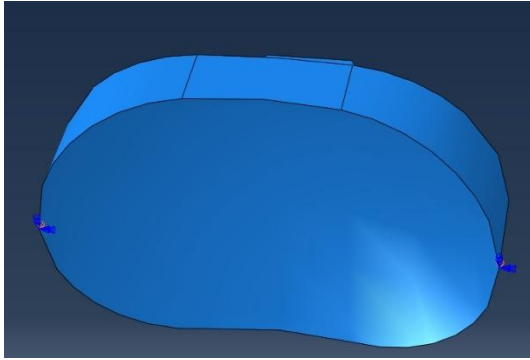


Figure II 2: First boundary condition applied to the model

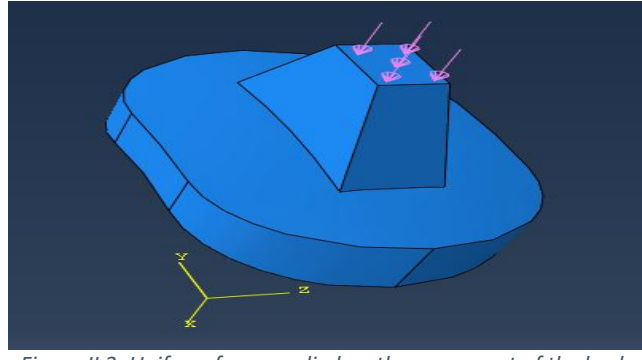


Figure II 3: Uniform force applied on the upper part of the keel

1st test: Model-1

- **Configuration:** A uniform force of 150N was applied to the keel surface. The support conditions were modeled by a blockage at two extreme points.

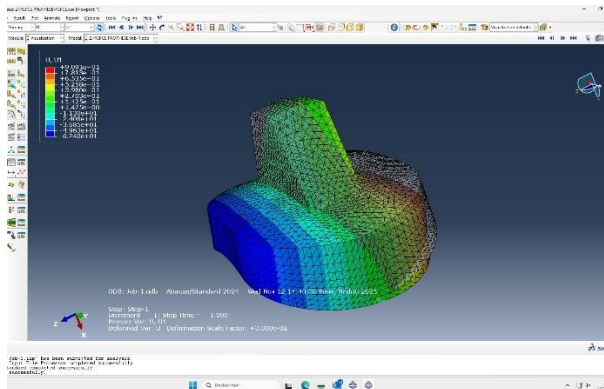


Figure II 4: displacement U1 following X in Abaqus (but following Y for the statement)

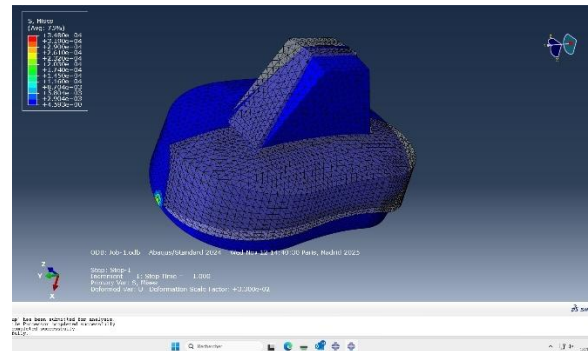


Figure II 5: Von mises

Comment: The sheath moves in rotation and the intensity of the arrow is too great, which highlights a problem (boundary condition, mesh) The model presented a rigid body rotation, indicating that the boundary conditions were insufficient and did not stabilize the part, not reflecting the experience.

2nd test : Model-1

- **Configuration:** The total force of 150N has been applied (instead of a uniform force). The CLs have been modified for a symmetry condition (XSYMM: U1 = UR2 = UR3 = 0) applied to all the points of the lower edges.

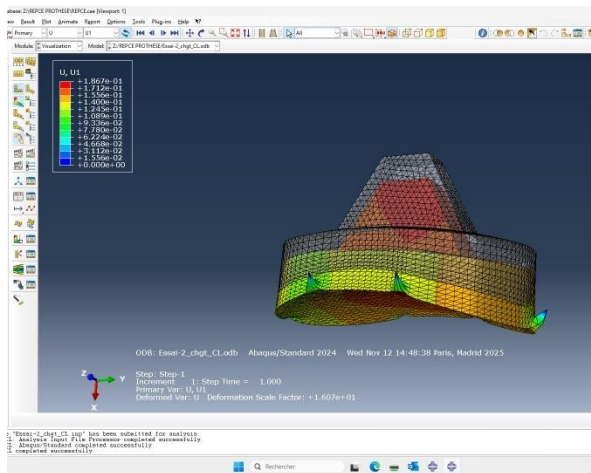


Figure II 6: displacement U1 following X in Abaqus (but following Y for the statement)

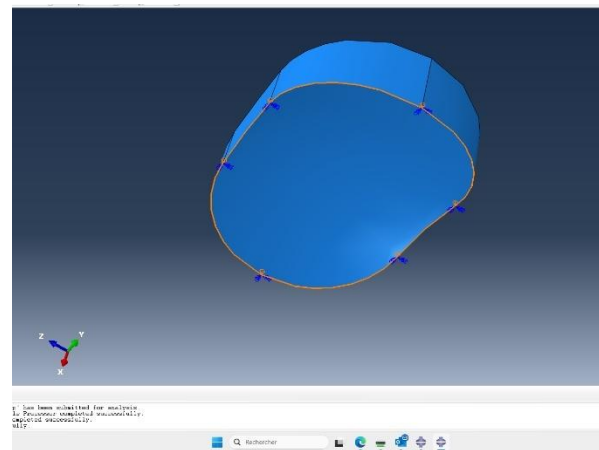


Figure II 7: boundary condition

Comment: Although this value is numerically close to the experimental result, i.e. 0.18mm (for an experimental value of 0.228 mm), the analysis of the deformation showed a physically unlikely behavior. The problem of boundary conditions persisted.

3rd test : Model-1

Configuration: we will set 2 boundary conditions instead of one:

- - 1 that blocks the ends (XSYMM)
- - 1 which blocks the lateral points (XASYMM)
- - 1 on the ends (XSYMM) and 1 on the lower lateral edges

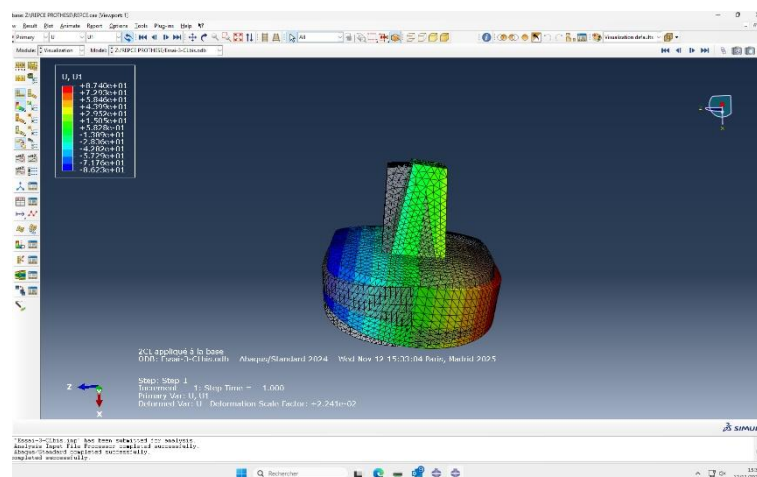


Figure II 8: displacement U1 following X in Abaqus (but following Y for the statement)

Comment: These configurations proved to be totally unrealistic compared to the physical assembly

Summary of the 3 trials: Although the boundary conditions reflect more reality in terms of deformation at the level of the part lying on the humeral head of the prosthesis, it results that there is always a problem with deformations. The mesh is good, so we should see if more precise conditions can allow us to find something reflecting reality. Indeed, the failure of previous tests

highlighted that the modeling of support could not be done by simply selecting native points or edges from the CAD geometry. The real support is done on the edge of the tray

4^e essai : Modèle-1

- **Configuration:** The approach has been fundamentally modified. Instead of selecting points, the extreme edges of the board (those coming into contact with the cubic support) have been partitioned

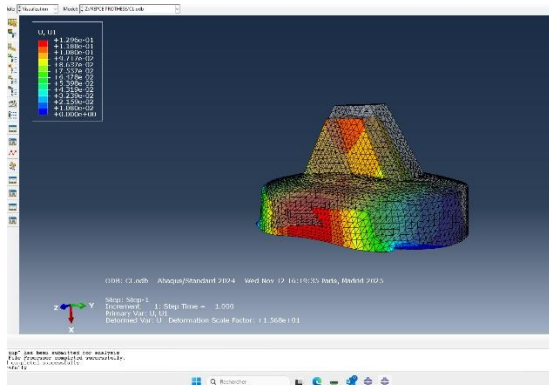


Figure II 9: displacement U1 following X in Abaqus (but following Y for the statement)

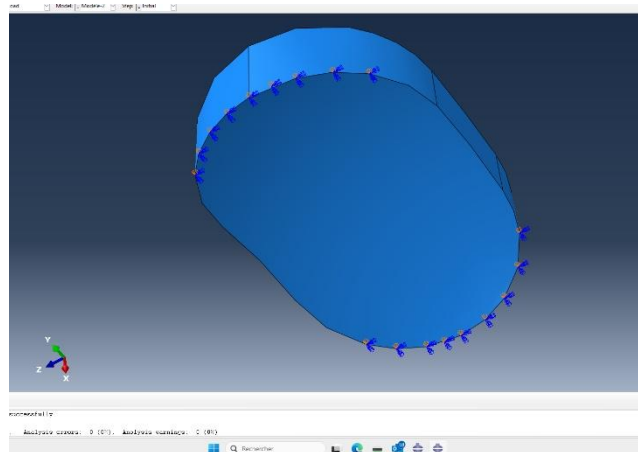


Figure II 10: Boundary conditions on the basis of ice

5^e essai : Modèle-2

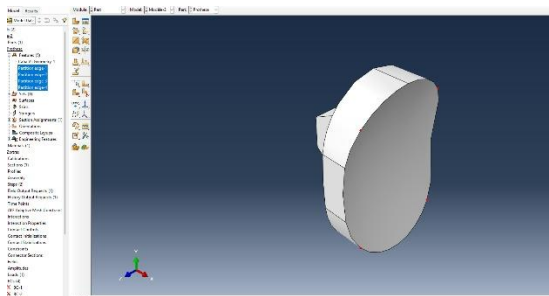


Figure II 11 : Partition de la zone de contact

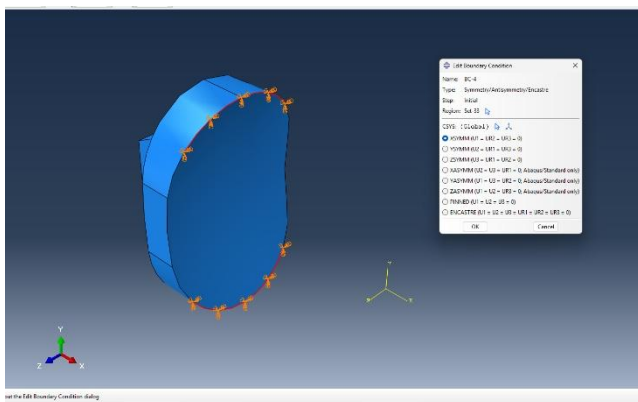


Figure II 12: boundary conditions at the level of the partitioned area

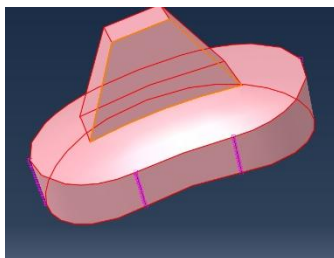


Figure II 13: regions selected to refine the mesh (in red/purple)

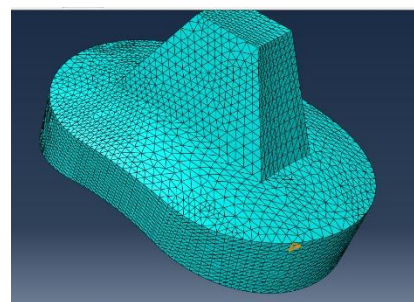


Figure II 14: mesh made

Configuration: This essay refines the approach of Essay 4. A symmetrical and distributed geometric partition was created on each side of the glenoid component, on the bearing surfaces of the plateau. In addition, a finer dege mesh on the edges rather than on the other parts has been established. We globally control the other edges (in this case: min/max = 0.2/1 (by elements). Then we control the average size of the elements present on the lateral edges (in this case: 0.2)

- **Analysis:** This final configuration is considered the most realistic. It correctly simulates the contacts with the support and should allow observing a symmetrical distribution of stresses and strains, in accordance with the axial loading.

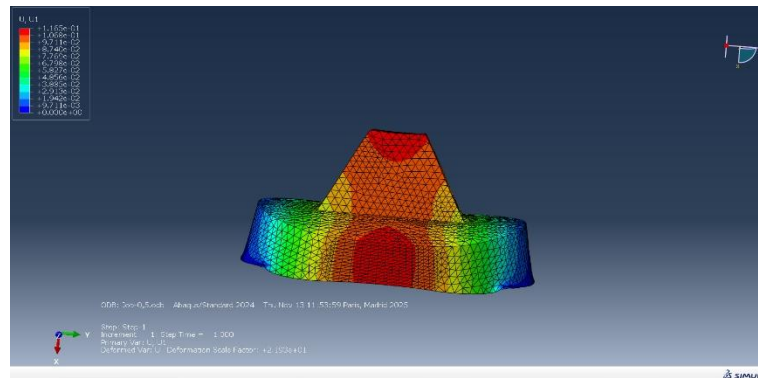


Figure II 15: displacement U1 following X in Abaqus (but following Y for the statement)

Comment: By looking at this model, we will be able to move on to the convergence of the mesh. However, the value is quite low for a fine mesh, so it will be necessary to reduce the contact surface. (see job-0.5)

6^e test : Model-2

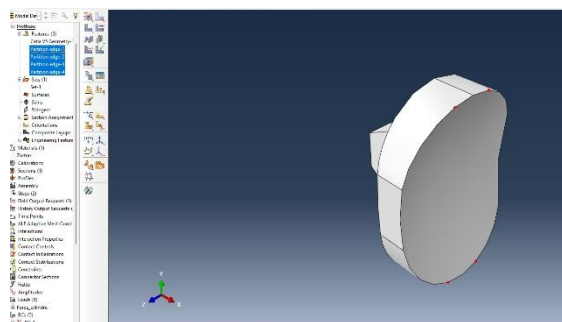


Figure II 15 : Partition améliorée de la zone de contact

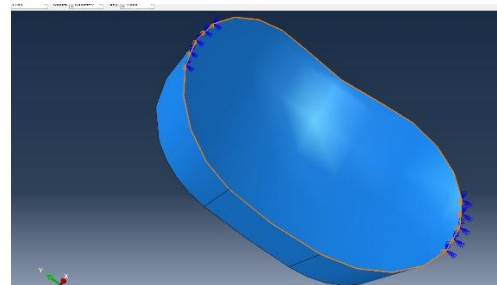


Figure II 16: Boundary conditions at the lower contact

Configuration: We had to refine the partition because the displacement was not significant enough for a mesh f. Indeed, the larger the contact surface, the smaller the U1 displacement will be. In this way, to better reflect the reality of the experience, we have reduced the contact surface.

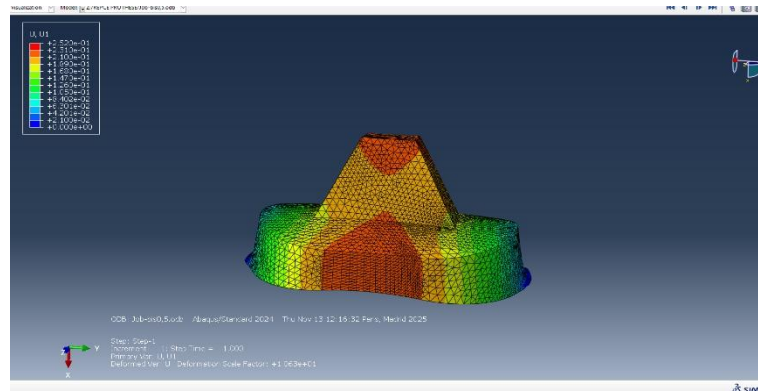
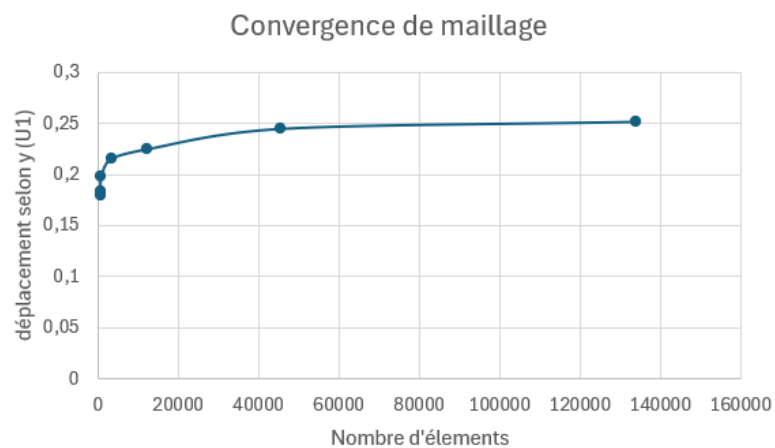


Figure II 17: displacement U1

Comment: For a fine mesh (0.5 and 0.5/1), we obtain a value almost similar to the experimental value with 2 hundredths of difference. Our mesh is a bit too thin, but it will just need to be configured as before.

II.3) mesh convergence



Our model is quite good since we can see an asymptote from a certain value of U1.

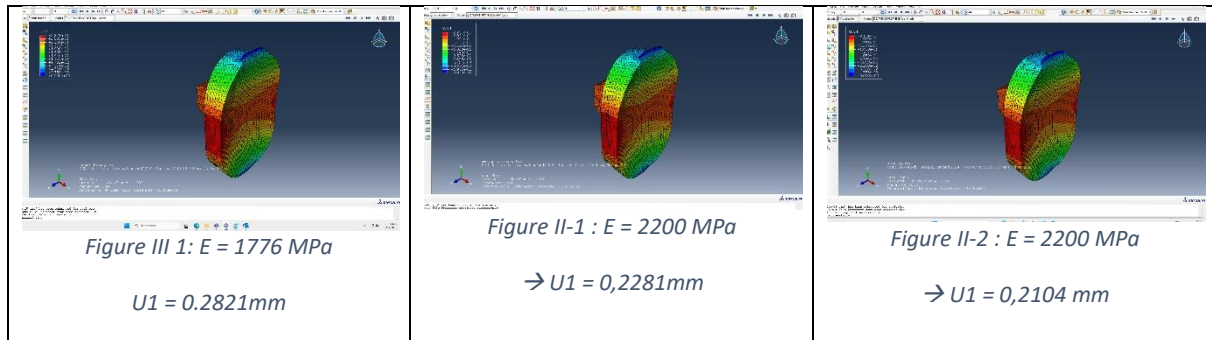
<u>mesh type</u>	<u>element size (min/max)</u>	<u>Overall size lateral edges</u>	<u>number of elements</u>	<u>U1(en mm)</u>
By region	6/6.5	1	522	0.1791
By region	5/5.5	1	576	0.1828
By region	4/4,5	1	761	0.1975
By region	2/2.5	1	3248	0.2142
By region	1/1,5	1	12258	0.2249
By region	0.5/1	1.5	45435	0.245
By region	0,5/1	0.5	133941	0.252

III) Modification of material properties

III.1) Modification of the modulus of elasticity

With a fine mesh and since we don't have a value that is not exact, we can play on the properties of the material.

Here are the observed Abaqus figures



job name	E (in MPa)	U1 (in mm)
Job-13	1779 (min value)	0.2821
Job-14	2385 (max value)	0.2104
Job-15	2200	0.2281

By increasing the modulus of elasticity, the material becomes more rigid, so a lesser displacement is observed: which is consistent with our values recorded on Abaqus. In the same way, with a lower modulus of elasticity that leads to greater displacement.

III.2) Influence des conditions limites pour obtenir la bonne valeur

Our boundary conditions are rather well placed so we are no longer obliged to play on this property.

IV) Conclusion

IV.1) Debate

The major issue and the main difficulty of this modeling did not lie in the choice of mesh or material properties, but in the definition of boundary conditions.

- **Initial failures (Trials 1-3):** The first attempts, based on blockages at extreme points or the application of symmetry conditions (XSYMM, XASYMM) on the native lower edges of the CAD, led to aberrant results. These models presented either rigid body movements (rotation) or physically inconsistent deformations, demonstrating a poor representation of contact with the support.
- **Solution by partitioning (Trials 4-6):** The analysis revealed that the real support was on the edge of the board. The solution was therefore to modify the geometry by partitioning the edges to create specific and symmetrical support surfaces. The application of CL on these new surfaces allowed to stabilize the model and to obtain a realistic physical behavior.

- **Final adjustment (Convergence and Material):** Once the CL correctly defined (Model-2), a mesh convergence study was conducted. With a fine mesh (133,941 elements), the displacement (0.252 mm) approached the target, but remained higher. The final step was to adjust the properties of the material, as planned. The initial Young's modulus (2.1 GPa) was slightly increased, while remaining within the valid range (1.78 - 2.38 GPa).

IV.2) Final conclusion

The validation of the EF model has been successfully completed. By applying an iterative approach centered on the definition of boundary conditions, a stable and physically coherent model was established (Model-2, Test 6) . By finally adjusting the modulus of elasticity to 2200 MPa, a realistic value for Zortrax resin, the model returned a displacement of 0.2281 mm. This result correlates almost perfectly with the target experimental value of 0.228 mm. The study therefore validates all the hypotheses (linear elastic model, quadratic tetrahedral mesh) and demonstrates the critical sensitivity of the model to precise modeling of contacts and supports.

